Experiment-8 Rotational Motion

To purpose of this experiment is to study the dynamics of a disk that rotates about an axis passing through its center and to calculate the angular acceleration, angular velocity and moment of inertia of that disk. Then we will investigate the conservation of the mechanical energy.

In this experiment we are going to study the rotational motion of a rigid body. A rigid body is an object with a definite shape that doesn't change, so that the particles composing it stay in fixed positions relative to one another. To study the rotational motion of a rigid body, we first need to introduce some new physical quantities and concepts that are essential to the description and understanding of this type of motion.

The first quantity that we are going to introduce is the **moment of inertia** of a rigid body. The moment of inertia (I) of an object is a measure of how much the object *resists* changing its rate of rotation about some axis. We know that in linear motion, if the same force were applied to objects with masses **m** and **M** (**M**>**m**), the one with smaller mass would have greater acceleration. In other words, the force would cause a greater change in the motion of the object with the smaller mass compared to that with the larger mass. So, the mass of an object is actually a measure of the inertia (*resistance*) that an object exhibits against changes in its motion and, the larger this mass is, the larger is the inertia.

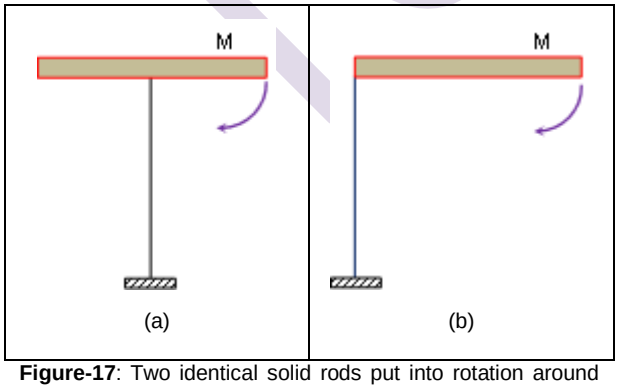


Figure-17: Two identical solid rods put into rotation around two different axes by the same force. It is easier to rotate the rod in (a) than it is in (b).

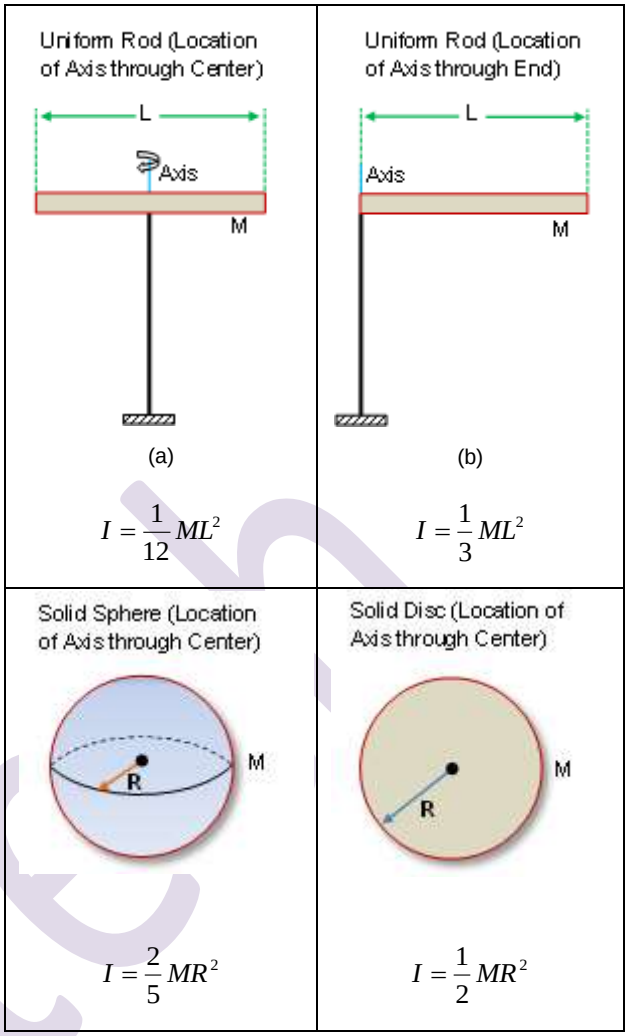


Figure-18: Moment of inertia for various objects of uniform composition.

Consider the situation described in the Figure-(17) where the two uniform rods with the same mass (**M**) are put into two different axes. It is clear that it is harder to rotate the rod in the Figure-(17b). That is, it has a larger moment of inertia compared to the other one, although they both have the same **mass M**. Therefore, the moment of inertia depends on the mass distribution with respect to the axis of rotation in addition to its mass. Figure-(18) gives the moment of inertia for some rigid bodies of uniform composition. Theoretically, for a disk of mass **M** and radius **R**, the moment of inertia through the center of symmetry is:

$$I = \frac{1}{2}MR^2 \qquad (67)$$

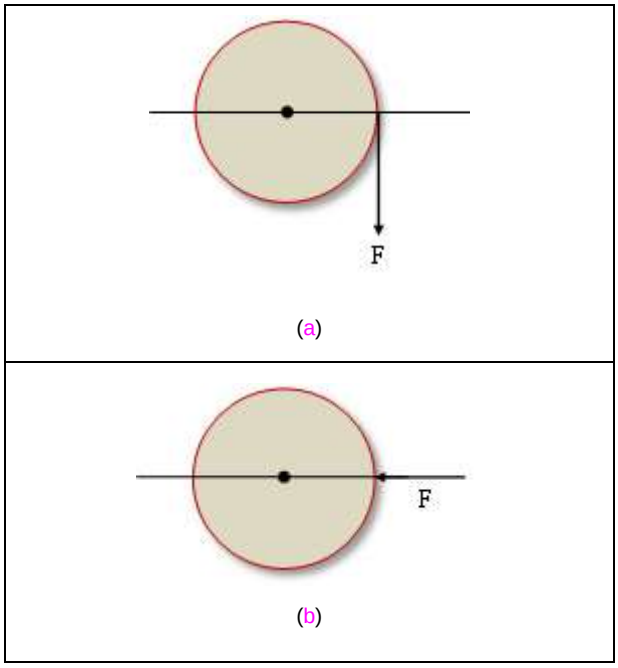


Figure-19: A net force acting on a **disk** that can rotate freely about an axis passing through its center (a). The same force acting as in (b) can not cause rotation.

We know that according to Newton's second law of motion when **a net force** acts on an object that is at rest, it puts it into motion. So, the net force acting tangentially on the disk causes it to rotate about the axis passing through its center. The **net force** (**F**) acting tangentially on the disk in the Figure-(19a) causes it to rotate about the axis passing through its center. However, when the same force acts on the same disk so that its line of action passes through its center as in the Figure-(19b), it can not rotate the disk. We see that a net force is not sufficient to cause a rigid body to rotate. So, we say that a rigid body will experience a change in its rotational motion if a net **torque** of that force about the rotation axis (or point) exists.

The **moment of inertia** of any object about any axis can be determined experimentally, by determining the net **torque** required to give the object an **angular acceleration** (α). The torque (**Γ**) of a force about a point or an axis is defined as:

$$\vec{\Gamma} = \vec{r} \times \vec{F} \qquad (68)$$

Here,

$\vec{r}$: The vector from the axis of rotation to the point where the force is applied,

$\vec{F}$: The applied force.

The torque causes an object to rotate. It depends on the net force and the point at which the force is applied relative to the axis of rotation. By the definition of the vector (cross) product, we have for the magnitude of the **torque**:

$$\Gamma = rF\sin\theta \qquad (69)$$

where θ is the angle between $\vec{r}$ and $\vec{F}$. This expression explains why the net force in the Figure-(19a) causes rotation, while that in the Figure-(19b) does not (that is, the angle θ is zero so Γ=0). Since *torque* is a distance times a force, it is measured in the units of newton-meter (Nm).

Net force acting on an object gives it a net linear acceleration but the torque acting on a rigid body gives it an **angular acceleration** (α) defined by:

$$\vec{\Gamma} = I\vec{\alpha} \qquad (70)$$

where I is the *moment of inertia* of the rigid body about the axis of rotation. The unit of moment of inertia is the kilogram-meter^2 (kgm^2).

As seen from the Equation-(70), to find the moment of inertia (rotational inertia) **experimentally**, a known **torque** (Γ) is applied to the object and the resulting **angular acceleration** (α) of the object is determined.

To describe rotational motion, we make use of ***angular*** quantities, such as angular velocity and angular acceleration. Now, let us define angular velocity (ω) and angular acceleration (α). These angular quantities are defined in analogy to the corresponding quantities in the linear motion and are chosen to describe the rotating object.

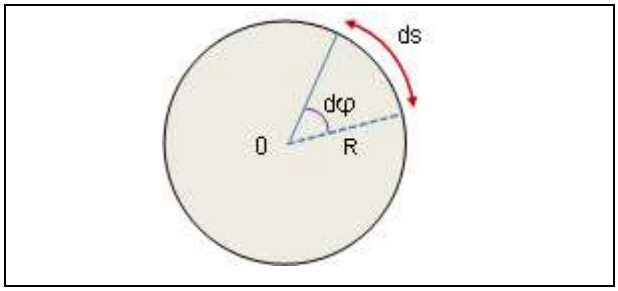


Figure-20: Change in the distance ds and the angle dφ in a time interval dt for a rigid disk rotating about an axis passing through its center.

Consider a rigid disk that is rotating about an axis passing through its center (**Figure-20**). We take the origin of the coordinates to be at the center of the disk. In a time interval **dt**, a point on the rim of the disk will travel a distance ds along the rim of the disk and the radius R will sweep an angle dφ. Each point of a rotating rigid object has, **at any instant**, a linear velocity and a linear acceleration. The magnitude of the **linear velocity** of the point on the rim of the disk will be:

$$v = \frac{ds}{dt} \tag{71}$$

Consequently, the **linear acceleration**, which is the rate of change of the velocity, will be:

$$a = \frac{dv}{dt} = \frac{d^2 s}{dt^2} \tag{72}$$

The **angular velocity** of the rotating disk is defined as the rate of change of the angle φ (measured in radians):

$$\omega = \frac{d\varphi}{dt} \tag{73}$$

Angular velocity has units of radians per seconds (rad/s) and note that all points in a rigid object rotate with the same angular velocity, since every position in the object moves through the same angle in the same time interval.

Angular acceleration is thus:

$$\alpha = \frac{d\omega}{dt} = \frac{d^2 \varphi}{dt^2} \tag{74}$$

Since ω is the same for all points of a rotating object, Equation-(74) tells us that α also will be the same for all points in a rigid body. With "ω" measured in radians per second and "t" in seconds, "α" has units of radians per **second squared** (**rad/s^2**). The length of the arc element *ds* is obviously;

$$ds = Rd\varphi \tag{75}$$

Therefore with ***R*** constant:

$$v = \frac{ds}{dt} = R\frac{d\varphi}{dt} \tag{76}$$

or;

$$\boxed{v = R\omega} \quad \textbf{(Linear Velocity)} \tag{77}$$

Consequently,

$$\boxed{a = R\alpha} \quad \textbf{(Linear Acceleration)} \tag{78}$$

where *R* is a fixed distance from the rotation axis and ω is given in rad/s.

From the above developments, we can relate the linear **quantities** at each point to the angular quantities of the rotating object as:

Linear and Rotational Quantities

Linear Motion	Rotational Motion
v	ω
a	α
m	I
$\vec{F} = m\vec{a}$	$\vec{\Gamma} = I\vec{\alpha}$
$v(t) = v_0 + at$	$\omega(t) = \omega_0 + \alpha t$
$K = \frac{1}{2}mv^2$	$K = \frac{1}{2}I\omega^2$

By comparing the expressions for *translational* and *rotational kinetic energies*, we see that angular velocity takes the place of linear velocity and moment of inertia (rotational inertia) takes the place of mass.

rentech

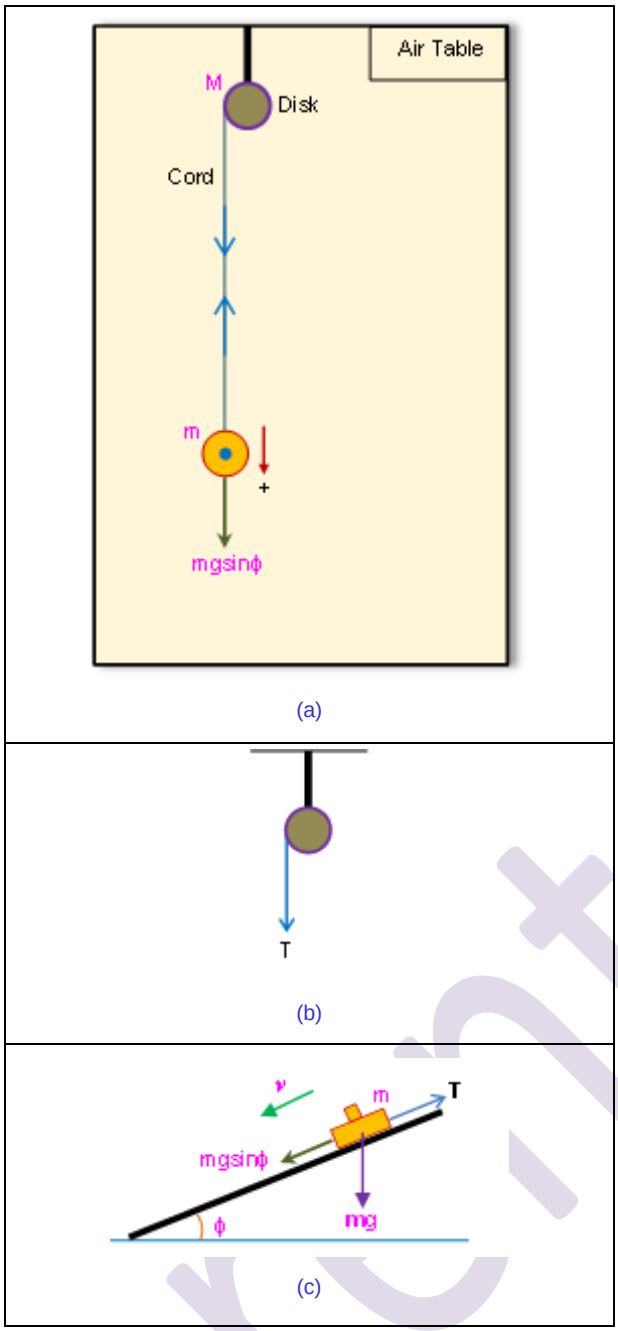


Figure-21: The experimental set-up (**a**). The applied force is the tension (**T**) in the string (**b**). Puck **m** performs linear motion downwards causing the disk **M** to rotate (**c**).

In this experiment, we are going to have the set up shown in the Figure-(21a), above. A solid disk of mass **M** that can rotate freely about an axis passing through its center is mounted to the upper edge of the inclined air table. A cord is wrapped around this disk and the free end of the cord is attached to a puck having the mass (**m**). As the system is released from the rest, the **suspending puck** (**m**) will accelerate down the plane of the inclined air table putting the **disk M** into **rotation**.

The tension **T** in the cord (string) will act as the force whose torque causes the disk to rotate (see Figure-21b). The forces acting on the suspended puck are shown in the Figure-(21c). Applying Newton's second law, we get:

$$mg\sin\phi - T = ma \quad \text{(Experimental)} \quad (79)$$

where ϕ is the angle of inclination of the air table, and a is the linear acceleration of the puck. The angular acceleration (α) of the rotating disk is related to the linear acceleration through:

$$a = R\alpha \quad \text{(Experimental)} \quad (80)$$

where R is the radius of the rotating disk. The torque caused by the tension (T) is given by:

$$\Gamma = RT = I\alpha \quad \text{(Experimental)} \quad (81)$$

where **I** is the **moment of inertia** of the disk. If the system is released from the rest, then at a later time **t**, the linear velocity of the puck will be given by:

$$v = at \quad \text{(Experimental)} \quad (82)$$

Similarly, the angular velocity of the rotating puck will be:

$$\omega = \alpha t \quad \text{(Experimental)} \quad (83)$$

Kinetic energy is the energy associated with the state of motion of an object. The kinetic energy of an object of mass (m) moving with speed (v) is defined as:

$$K = \frac{1}{2}mv^2 \quad \text{(Kinetic Energy)} \quad (84)$$

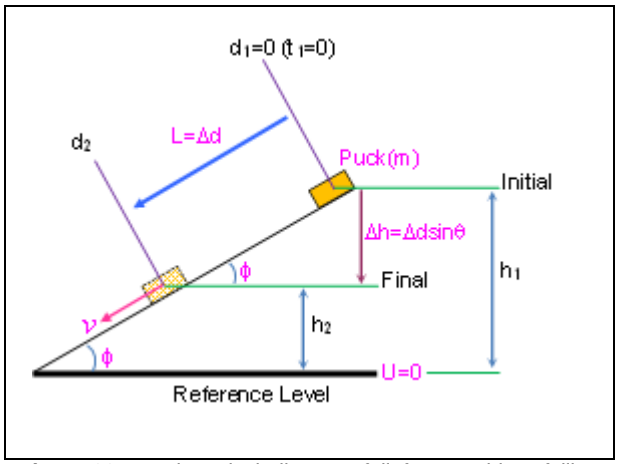


Figure-22: Total vertical distance fall for an object falling down the inclined plane.

The gravitational **potential energy** (**U**) of an object with mass (*m*) at any point of a **vertical height** (*y*) above a chosen reference level can be defined as:

$$U = mgy \qquad \text{(Potential Energy)} \qquad (85)$$

It is important to remember that potential energy is only defined relative to a location (that is, y=0), which can be chosen as the origin of a given coordinate system without affecting the motion of the mass.

As seen in the **Figure-(22)**, if an object of mass (**m**) slides or falls down a total **vertical distance** (**Δ*h***) along an **inclined plane**, its potential energy decreases by:

$$\Delta U = mg(\Delta h) \qquad (86)$$

during its downward motion. The lost potential energy is converted into other types of energy, such as **kinetic energy**. It is noticed that if the object is traveling a **distance** (**L**) along the inclined plane, the total *vertical distance* (**Δ*h***) will be:

$$\Delta h = L\sin\phi \qquad \text{(Vertical Distance)} \qquad (87)$$

where ϕ is the angle of the incline.

In the experiment, the **puck** has kinetic energy due to its translational (**linear**) motion and the **disk** has kinetic energy due to its **rotational** motion. In terms of moment of inertia, the **rotational** kinetic energy of an object rotating about a fixed axis is:

$$K = \frac{1}{2} I\omega^2 \qquad (88)$$

An object rotating about a fixed axis is said to have rotational kinetic energy.

The total kinetic energy of the system **at any instant** of time will thus be:

$$K = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2 \qquad \text{(Experimental)} \qquad (89)$$

The initial potential energy is transformed into kinetic energies of the **puck** and **disk**. As the **puck** falls down the inclined plane of the air table, its potential energy changes into **translational** and **rotational** kinetic energies of the **puck** (***m***) and **disk** (**M**), respectively.

Now, we can define a quantity **E**, called the **total mechanical energy** of the system as the sum of the kinetic energy (**K**) plus the potential energy (**U**) of the system at any moment:

$$E = K + U \qquad (90)$$

Letting **U** be the total potential energy of the system, conservation of energy (if the friction neglected) requires that:

$$\Delta K + \Delta U = 0 \qquad (91)$$

$$(K_2 - K_1) + (U_2 - U_1) = 0 \qquad (92)$$

$$\Delta U = -\Delta K \qquad (93)$$

where Δ means the change in energy. If the kinetic energy K increases, then the potential energy U must decrease by an equivalent amount to compensate. Thus the total mechanical energy K+U remains constant when the frictional forces do not act on the objects in the system. This is called the principle of the *conservation of mechanical energy* for conservative forces.

If no friction or non-conservative forces act, we can rewrite **Equation-(92)** as:

$$K_2 + U_2 = K_1 + U_1 \quad (94)$$

So, the quantity **E=K+U** at some time **1** is equal to **K+U** at any time **2**. So, when using the energy approach, we should first define what the initial and final states (the positions and velocities) of the system are. We can use the subscript **1** for initial position and the subscript **2** for the final position.

The **mechanical energy** of a system is the sum of its potential energy and the kinetic energy of the objects within it. Kinetic energy is the energy associated with the state of motion of an object. However, potential energy is only defined relative to a location, which can be chosen as the origin of a given coordinate system without affecting the motion of the mass. When the total mechanical energy of a system is conserved, we can relate the sum of kinetic energy and potential energy at **one instant** to that at **another instant** without considering the intermediate motion.

Due to **conservation of energy**, the **initial** potential energy should be converted completely into kinetic energy, which includes both translational and rotational energies. Therefore, at any instant of time after the system is put into motion, we have:

$$mg(\Delta d)\sin\phi = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2 \quad \text{(Experimental)} \quad (95)$$

Here, Δd is the change in the distance of the puck travelling down the inclined plane (that is, L=Δd). Before the experiment, you should first determine the system for which you apply energy conservation: the objects and the forces acting.

Equation-(95) is used to verify the conservation of mechanical energy for the system with a puck falling down and a rotating disk.

rentech

8.1. Experimental Procedures

In this part of the experiment, you will use an inclined air table. Therefore, first level off the air table, then bring it into the inclined position by using the block for inclination.

1. First, level off the air table and then tilt it to an inclined position with the angle (ϕ):
   - $\theta = 9^0$.

   by raising its backside and placing the block (the block for inclination) under it.

2. Attach one end of the cord to the rotating disk, **M** on the air table.
   - **2.1.** Attach the other free end of the cord to the suspended puck.
   - **2.2.** As **only** one puck is going to be used in this experiment, keep the other puck still at the lower corner of the air table by folding a piece of paper and putting it under the puck.

3. Adjust the suspended puck so that the rope is stretched, then pressing only the foot switch (P), allow it to fall down the inclined plane.
   - **3.1.** Observe the rotational motion of the disk.
   - **3.2.** Repeat this several times until you get a convenient motion.

4. Now, adjust the suspended puck again as in Step-2, and set the spark-timer frequency to:
   - f=10Hz.
   - **4.1.** Put the foot switches **S** (spark timer switch) and **P** (puck switch) on the top of each other and **press** them simultaneously.
   - **4.2.** Keep the switches depressed until the suspended puck reaches the bottom of the inclined plane.

5. **Turn off the spark-timer and then remove the pucks from the air table.**

6. Remove the data recording paper from the air table and examine the dots produced on it.
   - **6.1.** Is the trajectory of the puck a straight line?.
   - **6.2.** Are the dots evenly (equally) spaced or does the distance between successive dots increase with time?.
   - **6.3.** What kind of trajectory and dot spacing was expected?.
   - **6.4.** Does the data you obtained agree with what you expected?.
   - **6.5.** If you feel that your data is inconvenient, repeat the experiment again.

7. **Starting from the first dot**, number the dots as **0,1,2,3,4, and 5**.
   - **7.1.** Measure the time (**t**) and distance (x) of the each dot from the **dot-0** along with the corresponding errors.
   - **7.2.** Enter your data in the data **Table-(9)**.

8. Using the data in the data Table, **plot** x vs t^2 on a graph.
   - **8.1.** Take the **positive** x-direction in the direction of motion.
   - **8.2.** Now, **draw** the best line passing through the data values on the graph. Be careful with the units of vertical and horizontal axis of the graph.
   - **8.3.** Find the **slope** (s) of the best-fit line (straight line) and from this, determine the **linear acceleration** (a) of the **suspended puck**.
   - **8.4.** Remember that the **slope** of the x versus t^2 graph will be equal to:
     - $"\frac{1}{2}a"$.

   It is noticed that if your graph of **x-t^2** is **not** linear, you should repeat the experiment and again create a new "calculated column" in the data **Table-(9)** that will enable you to obtain a linear relationship.

9. Measure the mass (m) of the suspended puck.

rentech

10. Measure also the **radius** (R) of the **rotating disk** and then using the linear acceleration (a) in the equation:

- $a = R\alpha$

find the **angular acceleration** (α) of the rotating disk.

11. Then, obtain the **linear velocity** (v) of the suspended puck and **angular velocity** (ω) of the rotating puck at the end of the fifth interval (**at t=0.5s**).

12. Now, you will find both the **translational** and **rotational** kinetic energies of the suspending and rotating pucks.

12.1. Using these values of **velocity** (v) and **angular velocity** (ω), find the **total kinetic energy** (K_T) of the system *at this instant* (**at t=0.5s**).

12.2. Then, calculate the potential energy **lost** by the **suspending puck**.

12.3. Verify the **conservation of mechanical** energy for the system. Show your calculations.

12.4. Fill your data in the Table-(10), (11), (12), (13) and (14) with the correct **units**.

13. Using the **mass** (m) of the suspended puck in expression:

- $mg\sin\phi - T = ma$

find the tension (T) in the cord.

13.1. Using the equation:

- $\Gamma = RT$

determine the torque (Γ) of this tension.

13.2. Also find the ***experimental*** value of the moment of inertia (**I**) of the rotating disk by using:

- $\Gamma = I\alpha$

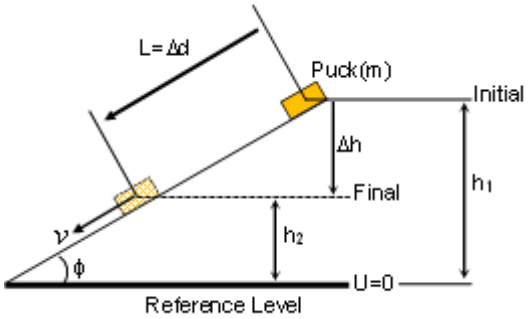
13.3. Now find the **theoretical** value of moment of inertia (**I**) of the disk by:

- $I = \frac{1}{2}MR^2$

13.4. Compare the **experimental** value of the moment of inertia (**I**) with the expected (**theoretical**) value.

13.5. Record you data in the Table-(15), (16) and (17).

14. Consider a puck having mass (**m**) sliding down an inclined plane of the air table as indicated by the diagram below.



Derive the equation for the potential energy **lost** in terms of the inclination angle (ϕ). Suppose the puck is released from rest so that the initial velocity v_0=0 at t=0.

Note that a reference point must be chosen to measure the puck's position vertical to the horizontal plane. For convenience, take h=0 and U=0. The angle (ϕ) of the inclined plane with the horizontal is also shown.

15. Write the expression describing the relationship between the net **torque** (Γ) acting on a rotating disk and its **angular acceleration** (α)?.

16. What is the relationship between the net **torque** acting on the disk and its **radius**?.

17. Write a general expression that represents the relationship between an object's **kinetic energy** due to its translational (linear) motion and its **velocity** (v).

18. Write a statement that gives the total kinetic energy of the mechanical system with a puck due to its **translational motion** and a disk due to its **rotational motion** at any instant of time.

19. Derive the equation describing **conservation of energy** (friction neglected) for an inclined air table system with a puck with linear motion and a disk with rotational motion at any instant of time after the system is put into motion.

rentech

8.2. LABORATORY REPORT

Rotational Motion

Table-9: Position and time measurements of the **suspended puck**.

Spark-Timer Frequency *f* (Hz)	Dot Number	Measured			Experimental	
		Time	Square of Time	Position	Slope	Acceleration
		t(sec)	$t^2(s^2)$	*x*(m)	$s(m/s^2)$	$a(m/s^2)$
10Hz	0	0	0	0	.....	.....
	1					
	2					
	3					
	4					
	5					

Record the time and position measurements of five data points of the suspended puck (m) relative to the dot 0.

Table-10: Translational kinetic energy **gained** by the **suspending puck**.

Suspended Puck		Measured	Calculated	Experimental
Mass	Linear Acceleration $(By\ Slope)$	Time	Linear velocity $(v = at)$	Translational Kinetic Energy $(K_1 = \frac{1}{2}mv^2)$
m(kg)	$a(m/s^2)$	t(s)	v(m/s)	K_1(J)
.....	.....	.....	.....	.....

Translational kinetic energy is determined at the instant of time t=0.5s corresponding to the dot-5 relative to dot-0.

rentech

Table-11: Angular acceleration and velocity of the **rotating disk**.

Rotating Disk		Experimental	Calculated	Experimental
Mass	**Radius**	**Linear Acceleration** $(By\ Slope)$	**Angular Acceleration** $(\alpha = \frac{a}{R})$	**Angular Velocity** $(\omega = \alpha t)$
M(kg)	R(m)	a(m/s^2)	α (rad/s^2)	ω (rad/s)
.....	.....	.....	.....	.....

Table-12: Rotational kinetic energy **gained** by the **rotating disk**.

Rotating Disk		Calculated	Calculated	Experimental
Mass	**Radius**	**Moment of Inertia** $(I = \frac{1}{2}MR^2)$	**Angular Velocity** $(\omega = \alpha t)$	**Rotational Kinetic Energy** $(K_2 = \frac{1}{2}I\omega^2)$
M(kg)	R(m)	I(kgm^2)	ω (rad/s)	K_2(J)
.....	.....	.....	.....	.....

Table-13: Total kinetic energy of the system **gained** by the **suspending puck (m)** and **rotating disk (M)**.

Suspended Puck	Rotating Disk	Experimental
Translational Kinetic Energy $(K_1 = \frac{1}{2}mv^2)$	**Rotational Kinetic Energy** $(K_2 = \frac{1}{2}I\omega^2)$	**Total Kinetic Energy** $(K_T = K_1 + K_2)$
K_1(J)	K_2(J)	K_T(J)
.....	.....	.....

Table-14: Potential energy **lost** by the **suspending puck**.

Puck	Acceleration due to Gravity	Measured	Measured	Experimental
		Change in the Distance	**Angle of the Incline**	**Potential Energy Lost** $(\Delta U = mg\Delta d\sin\phi)$
m(kg)	g(m/s^2)	Δd(m)	ϕ(deg)	ΔU(J)
.....	.....	.....	.....	.....

Table-15: Tension in the cord.

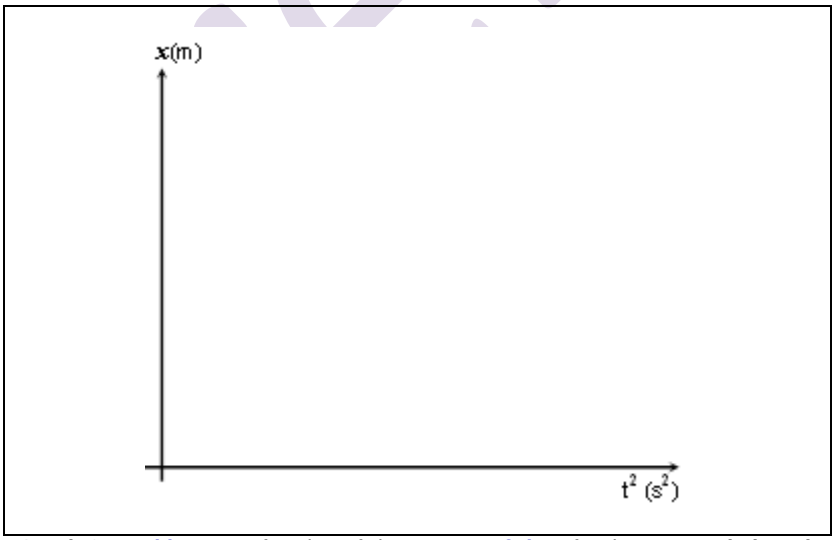
Puck	Acceleration due to Gravity	Measured	Experimental	Experimental
		Angle of the Incline	Acceleration ($By\ Slope$)	Tension ($mg\sin\phi - T = ma$)
m(kg)	g(m/s^2)	ϕ(deg)	a(m/s^2)	T(N)
.....	.....	.....	.....	.....

Table-16: Torque caused by the tension.

Rotating Disk	Experimental	Experimental
Radius	Tension ($mg\sin\phi - T = ma$)	Torque of the Tension ($\Gamma = RT$)
R(m)	T(N)	Γ(Nm)
.....	.....	.....

Table-17: Moment of Inertia of the disk.

Rotating Disk	Calculated	Experimental	Expected	Difference
Torque of the Tension ($\Gamma = RT$)	Angular Acceleration ($\alpha = \frac{a}{R}$)	Moment of Inertia ($\Gamma = I\alpha$)	Moment of Inertia ($I = \frac{1}{2}MR^2$)	Moment of Inertia ($\Delta I = I' - I$)
Γ(Nm)	α (rad/s^2)	I(kgm^2)	I′(kgm^2)	ΔI(kgm^2)
.....	.....	.....	.....	.....



Graph-6: Position as a function of the square of time for the **suspended puck** moving down the plane of the inclined air table with the linear acceleration.